

Exercise 1 – LTspice

- A. Simulate the I_{DS} vs. V_{GS} curve for an NMOS with $W=4\text{ }\mu\text{m}$, $L=2\text{ }\mu\text{m}$ and find the V_{th} . See file on Brightspace: **Week1_Ex1A.asc**
- Hint: plot $\text{sqrt}(I_{DS})$
- B. Find gm and r_o for $V_{GS}=1\text{ V}$ and $V_{DS}=1.8\text{ V}$:
- By computing the derivative of (I_{DS} vs. V_{GS}) and (I_{DS} vs. V_{DS})
 - Compare obtained gm with theoretical prediction
- C. Repeat point B for ($W=2\text{ }\mu\text{m}$, $L=1\text{ }\mu\text{m}$) and for ($W=0.4\text{ }\mu\text{m}$, $L=0.2\text{ }\mu\text{m}$). See: **Week1_Ex1C.asc**
- Compare the obtained values with theoretical predictions
 - Can you explain the eventual discrepancy?

Exercise 2 – LTspice

- A. Simulate the input-output characteristic of the simple amplifier in **Week1_Ex2A.asc**
 - Identify the regions of operations
- B. Find the bias conditions for maximum gain:
 - By simulation (Hint: compute derivative of I/O characteristic)
 - Compare the result to the prediction of the small-signal equivalent circuit
- C. Simulate the gain in the bias conditions as in point B but using AC analysis (see **Week1_Ex2C.asc**):
 - At a frequency $f = 1 \text{ kHz}$
 - At a frequency $f = 100 \text{ MHz}$
 - Can you explain the difference?

Exercise 3 – LTspice

- A. Simulate the input capacitance of an NMOS as a function of V_{GS} . Keep $4\text{ }\mu\text{m}$, $L=2\text{ }\mu\text{m}$, $V_{DS}=0.6\text{ V}$. See: **Week1_Ex3A.asc**
 - Can you explain the shape of the curve?
- B. Extract the various capacitive components of the small-signal equivalent circuit.